

Illustrative natural-capital valuation of an agri-environment heritage action: a methodological demonstrator

June 2026 - Kampakis and Co Ltd t/a The Tesseract Academy - lead: Dr S. Kampakis (CStat)

Illustrative only. This is a methodological demonstrator, not a commissioned appraisal. Unit values are indicative figures drawn from published sources to show how the method works; they are not validated estimates for any specific site and must not be cited as such. A full appraisal would derive site-specific values and confidence intervals.

1. Purpose

This note demonstrates how we would turn the multi-objective benefits of an AES heritage action into an indicative value-for-money statement, using the HM Treasury Green Book and Enabling a Natural Capital Approach (ENCA). It is published alongside our scoping-review protocol as a worked example of the value-for-money method.

2. Action and logic chain

We take action CHS5 / HEF6, Manage historic and archaeological features on grassland. By keeping buried features under permanent, undisturbed grassland (no cultivation, controlled stocking), the action avoids mechanical disturbance. That single management change generates a chain of ecosystem-service co-benefits alongside the primary heritage outcome.

Ecosystem service / benefit	Mechanism	Valuation route (Green Book / ENCA)
Soil organic carbon retention	Avoided soil disturbance retains carbon in archaeological soils	Non-traded carbon values (DESNZ); avoided-loss x carbon price
Water regulation	Permanent grass cover reduces run-off, erosion and sediment	Avoided water-treatment / sediment cost; ENCA databook transfer
Biodiversity (grassland)	Undisturbed semi-natural grassland supports species diversity	Habitat value transfer; biodiversity-unit equivalence
Heritage preservation	Non-renewable archaeological asset conserved in situ	Culture and Heritage Capital framework (DCMS/HE)
Access, landscape and wellbeing	Legible historic landscape supports access and engagement	Recreation / wellbeing value transfer; stated where not monetisable

3. Illustrative indicative valuation (per hectare per year)

The figures below are illustrative ranges assembled from published value sources to show the method. They are deliberately presented as ranges, not point estimates, and exclude any service where transfer is not defensible.

Co-benefit	Indicative annual value (GBP/ha/yr)	Basis (illustrative)
Soil carbon retention	10 - 60	Avoided soil-carbon loss at non-traded carbon values
Water regulation	5 - 40	Avoided sediment / treatment cost, value transfer
Biodiversity	15 - 90	Semi-natural grassland habitat value transfer
Recreation / wellbeing	Stated qualitatively	Reported in physical terms where monetisation is not defensible
Heritage (in situ asset)	Treated separately (CHC)	Culture and Heritage Capital, not double-counted with natural capital

Indicative result. Summed, the monetisable natural-capital co-benefits alone fall in an indicative range of roughly 30 to 190 GBP per hectare per year, before heritage and wellbeing value. Against scheme payment rates for the action, this illustrates how multi-objective accounting changes the value-for-money picture relative to a single-outcome view. The point of the demonstrator is the method and the honest treatment of uncertainty, not the specific numbers.

4. Treatment of uncertainty

- ranges, not point estimates; no false precision.
- services that cannot be defensibly monetised are reported in physical or qualitative terms, not forced into money.
- heritage value is handled through the Culture and Heritage Capital framework and kept separate from natural capital to avoid double-counting.
- a full appraisal would replace these transfers with site-specific evidence and sensitivity analysis.

5. Sources (illustrative value bases)

- HM Treasury, The Green Book (current edition).
- Defra, Enabling a Natural Capital Approach (ENCA) guidance and databooks.
- DESNZ, valuation of greenhouse gas emissions (non-traded carbon values).
- Defra / eftec, Valuing environmental impacts: guidelines for value transfer.
- DCMS / Historic England, Culture and Heritage Capital framework.
- ONS, UK natural capital accounts (2025).